SIMATS ENGINEERING

Department of Computer Science and Engineering

COMMON ASSIGNMENT REPORT

Library Book Reservation System using Zoho Creator (SaaS) with Big Data Analytics

Course Code: CSA1501
Course: Cloud Computing and Big Data Analytics

Submitted by

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Assignment scope: cloud/SaaS application design, Zoho Creator implementation evidence, library reservation workflow, big-data problem analysis, sample transaction dataset, and MapReduce-based reservation-frequency analytics.

CERTIFICATE AND DECLARATION

This report is submitted as the common assignment work for CSA1501 – Cloud Computing and Big Data Analytics. The assignment follows the prescribed requirement that teams independently design, develop, implement, test, analyze and document their solution, with evidence of implementation and learning outcomes.

Team Members

Name	Register Number	Role in the Assignment
FLORINA.K	192521479	Zoho Creator implementation and analytics integration
		Data model and testing.
VIJIYA PRIYA S.A	192524417	workflow analysis and dataset collection.
YASASWINI.N.R	192425119	report support documentation.

Declaration

We declare that this report represents our assignment work and that external sources, tools and course-provided material have been acknowledged. The report distinguishes the demonstrated Zoho Creator screens from the sample analytics dataset prepared for MapReduce experimentation.

Student Signature	Date
FLORINA.K	______________
	______________
VIJIYA PRIYA S.A	______________
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Source basis: The common-assignment instructions require a complete report covering problem formulation, objectives, requirements, relevant course concepts, design, algorithm/flowchart, implementation/tools, test cases, screenshots, results, analysis, SDG relevance, conclusion, contributions, references and individual reflection.

ACKNOWLEDGEMENT

We express our sincere gratitude to the Department of Computer Science and Engineering, SIMATS Engineering, for providing the common-assignment framework through which cloud computing and big-data concepts could be applied to a realistic library scenario.

We thank our course faculty for the guidance provided on Software as a Service, cloud access, virtualization concepts, big-data characteristics and MapReduce processing. The assignment encouraged us to move beyond a theoretical description and produce a working cloud application with forms, reports and sample records.

We also acknowledge Zoho Creator as the platform used to construct the cloud-based application interface and data-entry/reporting components. The screenshots included in this report document the application state available during development. The supporting screenshot evidence is retained in the accompanying screenshots PDF.

Finally, we thank our teammates and peers for discussions around database fields, reservation rules, testing scenarios and analytics design. The project helped us understand how a small information system can be extended into a cloud-accessible application and subsequently treated as a source of transaction data for big-data analysis.

Key learning areas

- Mapping a real-world library process into structured cloud forms and reports.
- Understanding why SaaS reduces the need for local infrastructure management.
- Designing validations and status transitions for reservations and book availability.
- Preparing a transaction dataset suitable for frequency analysis.
- Using mapper/reducer logic to convert reservation records into aggregated demand information.

The final report is organized to match the assignment question, assessment expectations and supplied report-subtopic structure while adapting the example domain from property management to library book reservation.

ABSTRACT

The Library Book Reservation System is a SaaS application hosted in the cloud and built on Zoho creator platform. In this system, we can manage the library books, members, reservations, waiting lists, book issues, returns, due dates and fines effectively.

The system enables student and faculty to register, search for books, know the availability of books, book or cancel book, wait in list and see transaction status. The librarian can create and update the records of books, handle reservations, give out and take in the books, track due dates and fines, as well as produce reports and dashboards.

Zoho Creator is selected as the SaaS platform as it supports cloud-based application building like forms, workflows, validations, reports, dash-boards, etc. The system is available over the cloud and the users do not need to worry about maintaining the application infrastructure.

The project also features Big Data Analytics based on Hadoop. Based on Big Data features such as volume, velocity, variety, veracity, privacy and scalability a sample library transaction dataset is constructed and analysed. An application on Hadoop MapReduce is designed to determine the reservation rate and to find the most reserved books or categories. The system is good example how the ideas of cloud computing,SaaS,virtualization,cloud collaboration, Big Data and MapReduce can be applied in the domain of library successfully and effectively.

For the large-data part, a toy library-transaction dataset is constructed with fields including transaction ID, member ID, book ID, title, category, date and status. The dataset serves to highlight issues of volume, velocity, variety, veracity/data quality, privacy and scalability. A MapReduce-style mapper emits (Book_ID, 1) for every reservation and the reducer sums over values for each key, yielding the number of reservations per book.

Keywords: Zoho Creator, SaaS, Cloud Computing, Library Management, Book Reservation, Hadoop, HDFS, MapReduce, Big Data Analytics.

1. INTRODUCTION

1.1 Background

A library contains a large collection of books that are used by students, faculty members and other authorized users. As the number of users and books increases, maintaining information manually becomes more difficult. A user may have to search through a catalogue or contact the librarian to find out whether a particular book is available. Similarly, librarians need to continuously update information related to reservations, book issues, returns and due dates.

A cloud-based library reservation system can make these activities easier by keeping the required information in one centralized application. Users can access the system through the internet without depending on a particular computer in the library. The librarian can update book and reservation information from the same application, allowing the latest information to be available to authorized users.

In this project, **Zoho Creator** is used to develop the Library Book Reservation System. The platform provides forms, reports, workflows and other application-building features that can be used to create the required library functions. The application follows the **Software as a Service (SaaS)** model because users access the developed application through the cloud rather than installing and maintaining the complete application on their own computers.

The project is also connected with Big Data Analytics. Library systems can generate a large amount of transaction information over time, including searches, reservations, issues, returns and cancellations. Analysing this information can help identify frequently reserved books and categories. Hadoop and MapReduce concepts are therefore included to demonstrate how library transaction data can be processed.

1.2 Need for the Project

The main reason for developing this system is to make the library reservation process more convenient and organized.

In a manual system, a user may need to ask the librarian about the availability of a book. When a popular book has already been issued, there may also be no simple way for the user to know their position in a waiting list. From the librarian's side, maintaining separate records for books, members, reservations and returns can increase the amount of repetitive work.

The proposed system addresses these issues by providing a centralized cloud application. Some of the important needs are:

- To provide a simple way to search for library books.

- To show the availability status of books.
- To allow students and faculty to reserve books.
- To support cancellation of reservations.
- To maintain a waiting list when a book is unavailable.
- To help librarians manage book and member information.
- To record issue, return, due-date and fine information.
- To generate useful reports and dashboard information.
- To analyse reservation data and identify books with higher demand.

The project is therefore useful from both an operational and analytical point of view. The cloud application supports everyday library activities, while Big Data processing can help understand usage patterns.

1.3 Problem Statement

In a conventional library system, information about books, members, reservations, issues and returns may be maintained manually or using separate records. This can result in delays, duplicate entries and difficulty in tracking the current availability of books.

When several users request the same book, the librarian must manage reservations and waiting lists while also keeping track of issue dates, due dates and returns. Therefore, a centralized system is required to manage these activities efficiently.

The proposed Library Book Reservation System using Zoho Creator provides a cloud-based solution in which students and faculty members can search for books, check availability, reserve or cancel books, join waiting lists and view reservation status. Librarians can manage books, reservations, issues, returns, due dates, fines and reports.

The system is further connected with Big Data Analytics to analyse library transaction data and identify frequently reserved books or categories.

1.4 Problem Formulation

The problem is formulated as the design and development of a cloud-based library reservation system that can manage different types of library information in a centralized manner.

The major information involved includes:

Book information – Book ID, ISBN, title, author, publisher, category and copies.

Member information – Member ID, name, department and contact details.

Reservation information – Reservation date, expiry date and reservation status.

Waiting-list information – Users waiting for unavailable books.

Issue and return information – Issue date, due date and return date.

Fine information – Fine amount and related transaction details.

The system should maintain accurate book availability when books are reserved, issued or returned. It should also provide suitable validation and workflows to reduce incorrect entries.

For the analytics component, library transaction data is prepared as a dataset and processed using MapReduce to calculate reservation frequency and identify highly demanded books or categories.

1.5 Objectives and Expected Outcomes

Objectives

The main objectives of the project are:

To develop a cloud-based Library Book Reservation System using Zoho Creator.

To maintain centralized records of books, members and library transactions.

To allow users to search for books and check their availability.

To provide book reservation and cancellation facilities.

To maintain a waiting list for unavailable books.

To help librarians manage issues, returns, due dates and fines.

To use validation and workflows to improve data management.

To generate reports and dashboards for library activities.

To prepare a sample library transaction dataset for Big Data analysis.

To implement MapReduce for calculating book reservation frequency.

Expected Outcomes

The expected outcomes of the project are:

A functional cloud-based library reservation application.

Organized book and member records.

Easier book searching and reservation.

Proper management of waiting-list entries.

Improved handling of issue and return transactions.

Reports and dashboards for monitoring library activities.

A prepared dataset for Big Data analysis.

Identification of frequently reserved books or categories using MapReduce.

Practical understanding of SaaS, cloud computing, Big Data and MapReduce concepts.

CHAPTER 2

REQUIREMENTS AND CLOUD SERVICE MODEL

2.1 Functional Requirements

The system should provide the following functions:

Register and manage library members.

Add and update book information.

Search books using title, author, category or ISBN.

Display book availability.

Allow users to reserve books.

Allow cancellation of reservations.

Maintain a waiting list for unavailable books.

Manage book issue and return transactions.

Calculate and maintain fine information.

Generate reports and dashboards.

Analyse reservation transaction data.

2.2 Non-Functional Requirements

The system should satisfy the following requirements:

Usability: The interface should be simple and easy to understand.

Reliability: Records should be stored consistently.

Security: Access should depend on the user's role.

Scalability: The system should support increasing books and users.

Availability: The cloud application should be accessible through the internet.

Performance: Search and record retrieval should be reasonably fast.

Maintainability: Forms and workflows should be easy to modify.

2.3 Hardware and Software Requirements

Hardware Requirements

Computer or laptop

Internet connection

Minimum 4 GB RAM

Standard keyboard and mouse

Software Requirements

Web browser such as Chrome or Edge

Zoho Creator

Internet connection

CSV/spreadsheet tool for dataset preparation

Python/Hadoop environment for MapReduce implementation

2.4 Constraints and Assumptions

Constraints

Internet access is required for the cloud application.

Zoho Creator features may depend on the available account plan.

The analytics demonstration uses a sample transaction dataset.

Large-scale real-time library data is not used in the prototype.

Assumptions

Each book has a unique Book ID.

Each member has a unique Member ID.

Users provide valid registration information.

Librarians maintain correct book and transaction records.

The sample dataset represents typical library transactions.

2.5 Cloud Service Models – IaaS, PaaS and SaaS

Cloud computing provides different service models.

IaaS (Infrastructure as a Service): Provides virtual machines, storage, networks and other infrastructure resources. The user manages the operating system and applications.

PaaS (Platform as a Service): Provides a platform for developing and deploying applications without managing the complete underlying infrastructure.

SaaS (Software as a Service): Provides ready-to-use software through the internet. Users can access the application without managing the underlying servers or software infrastructure.

2.6 Selection of SaaS and Zoho Creator

SaaS is suitable for this project because the Library Book Reservation System can be accessed through a web browser without requiring users to install or maintain the underlying infrastructure.

Zoho Creator provides a low-code environment for creating forms, workflows, reports and dashboards. It is therefore suitable for developing the proposed library application within a cloud environment.

CHAPTER 3

SYSTEM DESIGN AND PROPOSED SOLUTION

3.1 Proposed System

The proposed system is a cloud-based Library Book Reservation System developed using Zoho Creator. It provides a centralized platform for managing books, members, reservations, waiting lists and library transactions.

Students and faculty can access the system to search for books, check availability and reserve books. Librarians can manage books, members, issues, returns and reports.

The transaction data generated by the application can also be used for Big Data Analytics.

3.2 System Architecture

The system consists of the following major layers:

User Layer – Students, faculty and librarians access the application through a web browser.

Application Layer – Zoho Creator handles forms, workflows, validations and business operations.

Data Layer – Book, member, reservation, waiting-list and transaction records are stored in the cloud.

Analytics Layer – Transaction data is prepared and processed using Big Data and MapReduce concepts.

Reporting Layer – Reports and dashboards display useful library information.

Architecture Flow

Users → Zoho Creator Application → Cloud Database → Reports/Dashboard → Analytics

3.3 System Modules

1. Book Management

Stores Book ID, ISBN, title, author, publisher, edition, category, rack number and copy details.

2. Member Management

Maintains Member ID, member name, department, contact and member type.

3. Reservation Management

Allows members to reserve available books and maintains reservation dates and status.

4. Waiting List Management

Stores requests for books that are currently unavailable and maintains their queue position.

5. Issue and Return Management

Maintains issue date, due date, return date, fine amount and transaction status.

6. Analytics Module

Processes reservation data to identify frequently reserved books and categories.

3.4 Data Flow

The basic data flow is:

Member Registration → Book Search → Availability Check → Reservation → Issue → Return → Transaction Record → Analytics

When a user reserves a book, the system checks its availability. If a copy is available, the reservation is recorded. If no copy is available, the user can be added to the waiting list.

3.5 Database / Data Model

The main entities are:

Book

Member

Reservation

Waiting List

Issue and Return

A member can make multiple reservations. A book can have multiple reservation and transaction records. The Book ID and Member ID are used to connect related records.

3.6 User Roles and Cloud Collaboration

Student/Faculty

Search books

Check availability

Reserve books

Cancel reservations

View reservation status

Join waiting lists

Librarian

Add and update books

Manage members

Approve or manage reservations

Issue and return books

Maintain fines

View reports and dashboards

Role-based access ensures that users can access only the functions relevant to them.

3.7 Big Data Analytics Architecture

Library transactions generate data from reservations, issues and returns. The data is collected and prepared as a structured dataset.

The analytics process is:

Transaction Data → Data Cleaning → Dataset Preparation → MapReduce Processing → Aggregated Results → Reports

MapReduce can group transactions by Book ID or category and calculate reservation frequency.

3.8 Virtualization and Cloud Infrastructure

Cloud infrastructure uses virtualized resources to provide computing, storage and application services. In this project, users do not directly manage the physical infrastructure. Zoho Creator provides the cloud environment required to access and operate the application.

CHAPTER 4

IMPLEMENTATION USING ZOHO CREATOR

4.1 Zoho Creator Implementation Procedure

The application is implemented using the following steps:

Open Zoho Creator and sign in.

Create a new application for the Library Book Reservation System.

Create the Book Details form.

Add the required book fields.

Create the Member Registration form.

Create the Book Reservation form.

Create the Issue and Return form.

Create the Waiting List form.

Configure relationships between related records using suitable lookup fields.

Add field validations.

Create workflows for reservations, issues and returns.

Create reports for different library activities.

Design a dashboard containing important library statistics.

Add sample records and test the application.

4.2 Validation and Workflows

Validation is used to prevent incorrect data entry.

Examples include:

Book ID must be unique.

Member ID must be unique.

Required fields cannot be left empty.

Number of copies must be a positive value.

Available copies cannot exceed total copies.

Fine amount cannot be negative.

Reservation dates should be valid.

A reservation should be allowed only when a copy is available.

Workflows are used to automate operations such as updating reservation status, updating available copies after issue or return, maintaining waiting-list information and notifying users about reservation-related changes.

4.3 Reports and Dashboard

The application provides reports such as:

Available Books

Reserved Books

Issued Books

Overdue Books

Waiting List

Most Reserved Books

Book Category Summary

The dashboard provides a quick view of important information such as total books, available copies, active reservations, issued books and overdue books.

4.4 Tools and Environment Used

The main tools used are:

Zoho Creator – Application development

Web Browser – Accessing the cloud application

Internet – Cloud connectivity

CSV/Spreadsheet – Dataset preparation

Python/Hadoop – MapReduce implementation and analytics

4.5 Implementation Screenshots

The following screenshots should be included in the final report:

Zoho Creator application home page.

Book Details form.

Member Registration form.

Book Reservation form.

Issue and Return form.

Waiting List form.

Validation/error message.

Workflow configuration.

Book availability report.

Reservation report.

Dashboard showing library statistics.

Sample records/output.

CHAPTER 5

BIG DATA ANALYTICS

5.1 Library Transaction Dataset

A sample library transaction dataset is prepared to demonstrate Big Data Analytics. Each record represents a library activity such as a reservation, issue or return.

The dataset contains information such as:

Transaction ID

Book ID

Book Title

Member ID

Category

Transaction Type

Transaction Date

Status

The dataset can be stored in CSV format for analysis.

5.2 Dataset Fields and Preparation

Before analysis, the dataset is checked for missing values, duplicate records and inconsistent entries.

The main preparation steps are:

Collect transaction records.

Remove duplicate records.

Check missing values.

Standardize dates and categories.

Verify Book IDs and Member IDs.

Store the cleaned data in CSV format.

Use the cleaned dataset for MapReduce processing.

5.3 Big Data Characteristics

Volume

As the number of books and users increases, the number of library transactions also increases. Over a long period, these records can form a large dataset.

Velocity

Transactions are generated continuously whenever users reserve, issue or return books. Therefore, new data can be generated frequently.

Variety

Library data may include structured fields such as IDs, dates and status, along with reports and other forms of information.

Data Quality

Incorrect, duplicate or incomplete records can affect analysis. Therefore, data cleaning is required before processing.

Privacy

Member information such as contact details should be protected and accessed only by authorized users.

Scalability

The system should be able to handle an increasing number of books, members and transactions without major changes to the overall design.

5.4 Data Analysis and Observations

The transaction dataset can be analysed to identify:

Most frequently reserved books.

Most demanded categories.

Number of reservations over time.

Number of issued and returned books.

Books with high demand.

CHAPTER 6

MAPREDUCE IMPLEMENTATION

6.1 MapReduce Approach

MapReduce is a distributed data-processing model used to process large datasets.

For the library system, MapReduce is used to calculate how many times each book has been reserved.

The process contains two main stages:

Map: Reads each transaction and produces a key-value pair such as (Book ID, 1) for a reservation.

Reduce: Groups the values belonging to the same Book ID and adds them to calculate the total reservation count.

6.2 Algorithm

Read the library transaction dataset.

Select reservation transactions.

Extract the Book ID.

Generate (Book ID, 1) during the Map phase.

Group records with the same Book ID.

Add the values during the Reduce phase.

Produce the reservation count for each book.

Identify highly reserved books.

6.3 Pseudocode

```
START

Read transaction dataset

FOR each transaction

  IF transaction type is "Reservation"

     Emit(Book_ID, 1)
```

END IF

END FOR

FOR each Book_ID

Add all emitted values

Store reservation count

END FOR

Display Book_ID and reservation count

STOP

6.4 Flowchart

START

↓

Read Dataset

↓

Select Reservation Records

↓

Map: Generate (Book ID, 1)

↓

Shuffle and Group by Book ID

↓

Reduce: Add Counts

↓

Generate Reservation Frequency

↓

Display Results

↓

END

6.5 MapReduce Source Code

A simple Python implementation for demonstrating the MapReduce logic is:

```
from collections import defaultdict

import csv

reservation_count = defaultdict(int)

with open("library_transactions.csv", "r") as file:

  reader = csv.DictReader(file)

  for row in reader:

     if row["Transaction_Type"].lower() == "reservation":

        reservation_count[row["Book_ID"]] += 1

for book_id, count in reservation_count.items():

  print(book_id, count)
```

This program reads the transaction dataset, selects reservation records and counts reservations for each book.

6.6 Input and Output

Input

The input is a CSV file containing library transaction records.

Example:

```
Transaction_ID,Book_ID,Transaction_Type

T001,B101,Reservation

T002,B102,Issue

T003,B101,Reservation

T004,B103,Reservation
```

Output

B101 2

B103 1

The output represents the number of reservations for each book.

6.7 Analysis of Results

The output helps identify books with higher reservation frequency. A book with a larger reservation count can be considered highly demanded.

This information can help librarians decide which books require additional copies or better availability management.

CHAPTER 7

TESTING AND VALIDATION

7.1 Testing Methodology

Testing is performed to verify that the application works according to the requirements. Different forms, validations, reservation operations and reports are tested using sample records.

The main testing areas are:

• Form validation

• Book search

• Book reservation

• Waiting-list management

• Issue and return

• Fine information

• Reports and dashboard

• MapReduce processing

7.2 TEST CASES

The following test cases were designed and executed to verify the functionality of the system.

Test Case ID	Test Case Description	Input / Action	Expected Result	Actual Result	Status
TC01	Add a new book with valid details	Enter all book details and click Save	Book record should be saved successfully	Book record saved successfully	Passed
TC02	Add a book with duplicate Book ID	Enter existing Book ID and click Save	System should show error and reject the entry	Error shown: "Book ID already exists"	Passed
TC03	Search for an available book	Enter book title/author and click Search	Matching books should be displayed	Matching books displayed correctly	Passed
TC04	Reserve an available book	Select available book and click Reserve	Reservation should be created and status updated	Reservation created successfully	Passed
TC05	Reserve an unavailable book	Select unavailable book and click Reserve	User should be added to waiting list	User added to waiting list with position	Passed
TC06	Enter invalid number of copies	Enter negative value in Number of Copies field	Validation error should be displayed	Error shown: "Enter valid number"	Passed
TC07	Issue a reserved book	Select reservation and click Issue	Issue record created and availability updated	Issue record created successfully	Passed
TC08	Return an issued book	Select issued book and click Return	Return date recorded and availability updated	Return recorded successfully	Passed
TC09	Generate reservation report	Select report and provide date range	Reservation report should be generated	Report generated successfully	Passed
TC10	Run MapReduce program for reservation frequency	Execute MapReduce code on dataset	Reservation frequency should be generated	Reservation frequency output generated	Passed

7.3 Expected and Actual Results

The expected results are compared with the actual results obtained during testing. The tested functions produce the intended results for the sample records.

7.4 Validation and Observations

Testing confirms that:

• Required fields are validated.

• Book and member records are maintained.

• Reservations are recorded correctly.

• Unavailable books can be added to the waiting list.

• Issue and return records can be maintained.

• Reports display relevant information.

• MapReduce produces reservation frequency results.

7.5 Execution Screenshots / Outputs

Screenshots should be included for:

• Successful book entry.

• Successful member registration.

• Successful reservation.

• Validation error.

• Waiting-list entry.

• Issue and return operation.

• Generated reports.

• Dashboard.

• MapReduce input and output.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Results

The developed system provides a centralized cloud-based platform for managing library books and reservations.

The main results are:

- Book records can be maintained digitally.
- Members can search and reserve books.
- Waiting lists can be maintained.
- Issue and return records can be stored.
- Reports and dashboards provide useful information.
- Transaction data can be analysed using MapReduce.

8.2 Advantages

The major advantages are:

- Easy access through the cloud.
- Reduced manual record keeping.
- Centralized data management.
- Faster reservation processing.
- Better tracking of book availability.
- Automated workflows.
- Useful reports and dashboards.
- Supports data-driven library decisions.

The proposed system provides better organization and accessibility compared with a completely manual process.

8.3Trade-offs and Justification

Using a cloud-based SaaS platform reduces the need to manage servers and application infrastructure. However, the system depends on internet connectivity and the features available in the selected SaaS platform.

Zoho Creator is selected because it provides a practical environment for developing forms, workflows, reports and dashboards without requiring complete infrastructure management.

8.4 Limitations

The main limitations are:

- The prototype uses a sample dataset.
- Internet connectivity is required.
- Advanced analytics may require additional tools.
- Large-scale real-time processing is not fully demonstrated.
- Some cloud features may depend on the Zoho account plan.

8.5 Possible Improvements

Future improvements can include:

- Mobile application support.
- Barcode or QR-based book identification.
- Email or mobile notifications.
- Advanced recommendation systems.
- Real-time analytics dashboards.
- Integration with larger library databases.
- Predictive analysis for book demand.

CHAPTER 9

BROADER CONSIDERATIONS

9.1 SDG Relevance

The project supports the following Sustainable Development Goals.

SDG 4 – Quality Education

The system improves access to library resources by making book searching and reservation easier for students and faculty.

SDG 9 – Industry, Innovation and Infrastructure

The project demonstrates the use of cloud computing, SaaS and Big Data technologies to develop a digital library service.

SDG 12 – Responsible Consumption and Production

Digital record management reduces dependence on paper-based library records and supports more efficient use of library resources.

9.2 Practical Benefits

The proposed system provides practical benefits to both users and librarians.

Students and faculty can access book information and reservation services more conveniently. Librarians can manage records, transactions and reports from a centralized system.

Analytics can also help librarians identify books with high demand and make better decisions about resource allocation.

9.3 Data Privacy and Responsible Data Usage

The system contains member information and transaction records. Therefore, data should be accessed only by authorized users.

Role-based access should be used to protect sensitive information. Only necessary member information should be collected, and records should be handled responsibly.

The analytics dataset should also avoid unnecessary personal information wherever possible.

CHAPTER 10

CONCLUSION AND COURSE OUTCOME

10.1 Conclusion

The Library Book Reservation System using Zoho Creator provides a cloud-based approach for managing library activities. The system supports book management, member registration, reservations, waiting lists, issue and return operations, reports and dashboards.

Using Zoho Creator demonstrates the SaaS model by providing application services through the cloud without requiring users to manage the underlying infrastructure.

The project also demonstrates Big Data concepts by preparing library transaction data and applying MapReduce to calculate reservation frequency. The resulting information can help identify highly demanded books and support better library management.

Overall, the project combines Cloud Computing and Big Data Analytics concepts to develop a practical solution for library resource management.

10.2 CO Mapping

Course Outcome	Project Contribution
CO1	Understanding and application of SaaS using Zoho Creator
CO2	Understanding cloud infrastructure and virtualization concepts
CO3	Cloud-based application access, workflows and collaboration
CO4	Analysis of Big Data characteristics in library transactions
CO5	MapReduce implementation for reservation frequency

10.3 Individual Contribution of Group Members

The group members contributed to different activities of the project, including:

• Requirement analysis

• Zoho Creator application development

• Form and database design

• Workflow configuration

• Dataset preparation

• MapReduce implementation

• Testing

• Documentation

• Presentation preparation

Each member contributed to the development and validation of the project.

10.4 Individual Reflection

Working on this project provided practical knowledge of cloud computing and Big Data Analytics. The development of the Library Book Reservation System helped in understanding how a SaaS platform can be used to create forms, manage records, automate workflows and generate reports.

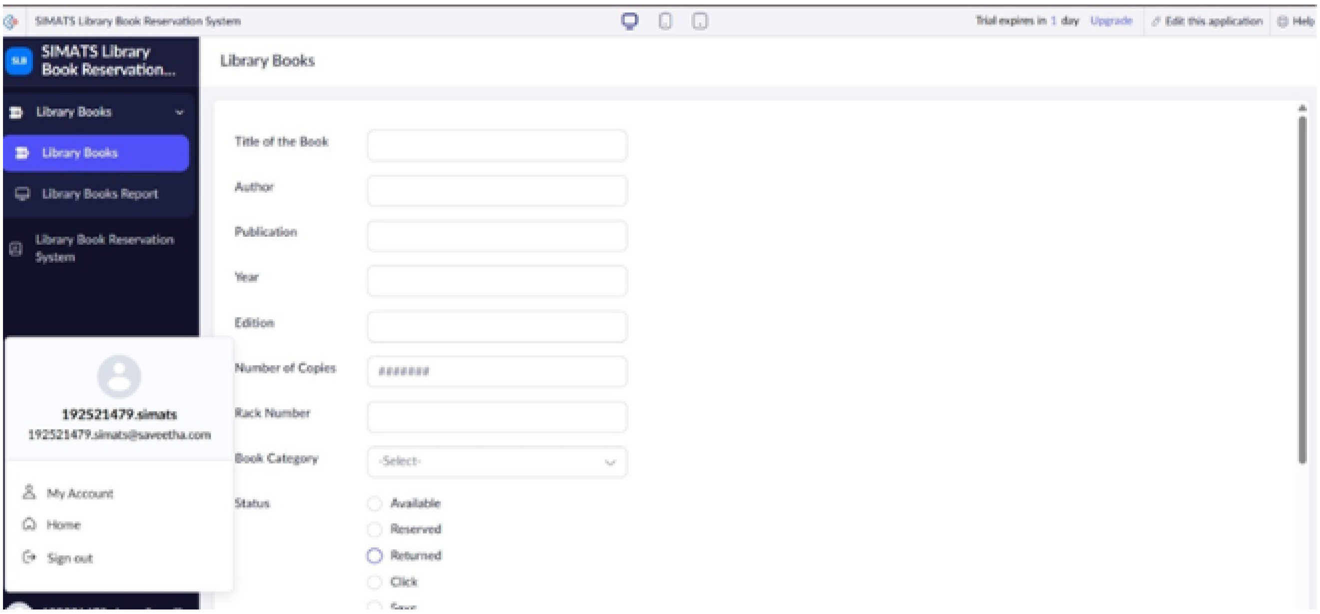
One of the important challenges was designing the data structure so that book, member and transaction information could be maintained consistently. Preparing the transaction dataset also helped in understanding the importance of data cleaning and organization before analysis.

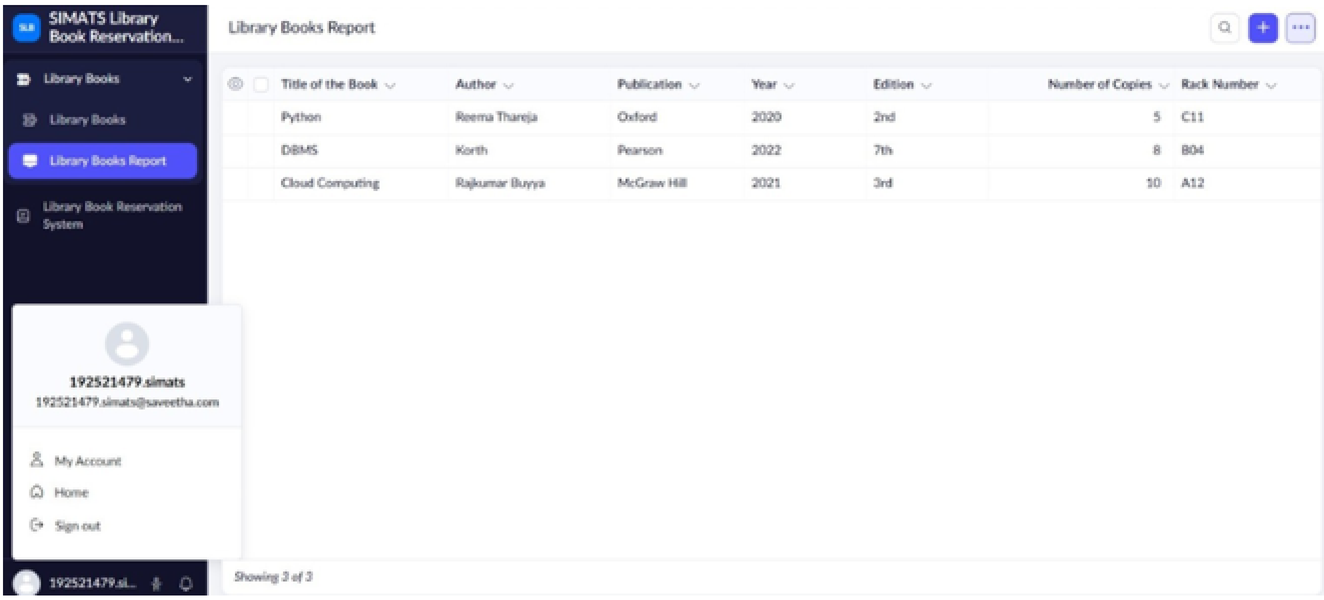
Implementing the MapReduce logic provided an understanding of how large transaction datasets can be processed by dividing the task into mapping and reducing stages. The project also improved practical knowledge of cloud collaboration, data management and analytical thinking.

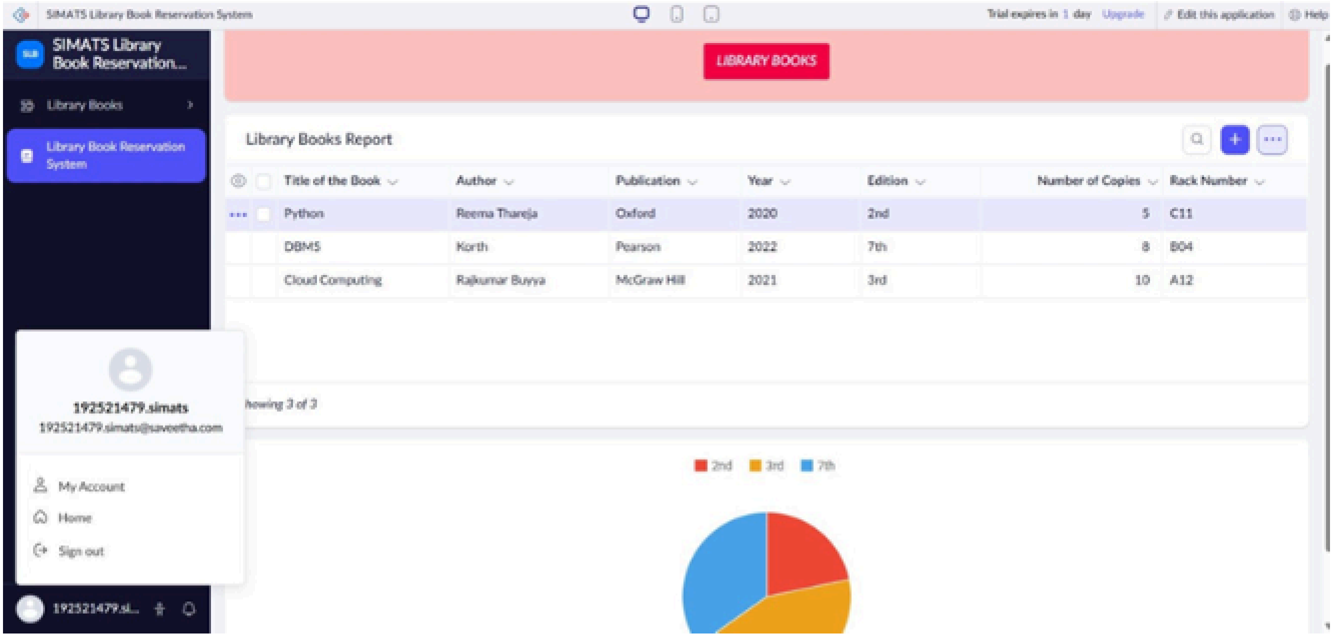
The project is related to SDG 4 through improved access to educational resources, SDG 9 through the use of cloud and data technologies, and SDG 12 through digital and efficient resource management.

Overall, the project helped connect theoretical concepts from Cloud Computing and Big Data Analytics with a practical library management application.

IMPLEMENTATION EVIDENCE






SIMATS Library Book Reservation System
Trial expires in 1 day Upgrade
Edit this application
Help
SIMATS Library Book Reservation...
Library Books
Library Book Reservation System
LIBRARY BOOKS
Library Books Report
Title of the Book
Author
Publication
Year
Edition
Number of Copies
Rack Number
Python
Reema Thareja
Oxford
2020
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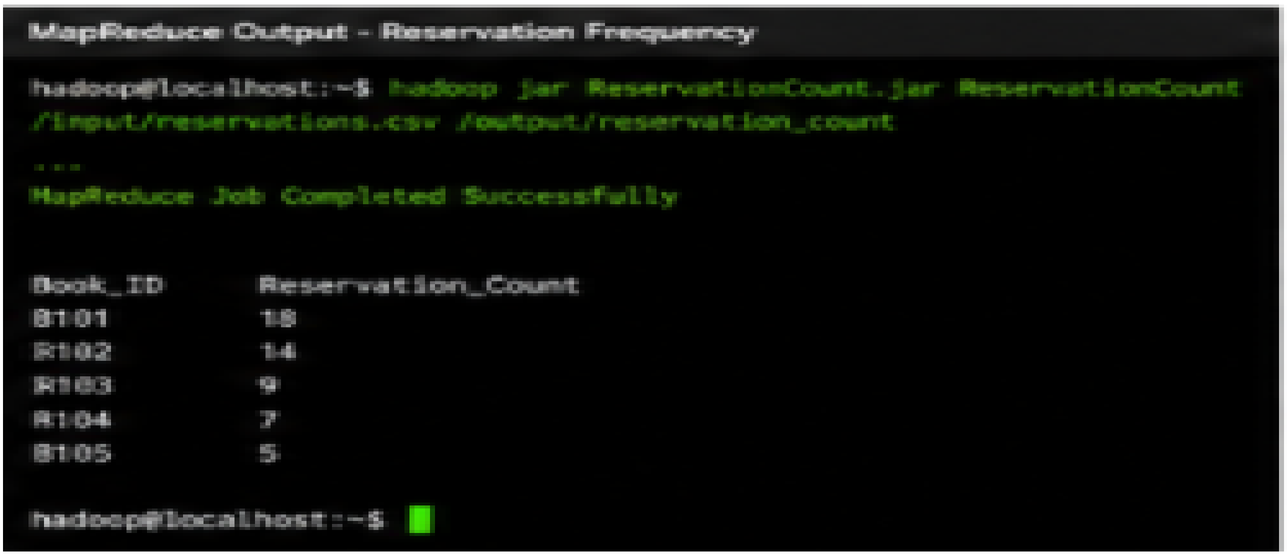
MapReduce Output - Reservation Frequency
hadoop@localhost:~$ hadoop jar ReservationCount.jar ReservationCount /input/reservations.csv /output/reservation_count
...
MapReduce Job Completed Successfully
Book_ID Reservation_Count
B101 18
R102 14
R103 9
R104 7
B105 5
hadoop@localhost:~$

MAPREDUCE SOURCE CODE AND ANALYTICS OUTPUT

The following compact Python scripts implement the mapper and reducer logic used for the sample CSV dataset. They follow the key-value idea of MapReduce and can also be adapted to Hadoop Streaming by using standard input/output.

mapper.py

```
#!/usr/bin/env    python3
import sys, csv

reader = csv.DictReader(sys.stdin) for row
in reader:
    print(f"{row['Book_ID']}\t1")
```

reducer.py

```
#!/usr/bin/env    python3
import sys

current_key = None total
= 0
for line in sys.stdin:
    key, value = line.strip().split("\t") value =
    int(value)
    if current_key == key: total
        += value
    else:
        if current_key is not None:
            print(f"{current_key}\t{total}")
        current_key = key total
        = value

if current_key is not None:
    print(f"{current_key}\t{total}")
```

For local testing, the mapper output can be sorted by key and piped into the reducer. The generated result file contains Book_ID, Book_Title and Reservation_Frequency. In the supplied sample, B001 is the highest-frequency book with 13 reservations.

Source files delivered with this report package: library_transactions.csv, mapper.py, reducer.py and mapreduce_book_frequency.csv.

References

[1] P. Mell and T. Grance, The NIST Definition of Cloud Computing, NIST Special Publication 800-145, National Institute of Standards and Technology, 2011. Available: https://doi.org/10.6028/NIST.SP.800-145. Accessed Sep. 2, 2026.

[2] Zoho Corporation, Understand Application – Zoho Creator Help. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/creating-applications/understand-applications/articles/understand-application. Accessed Sep. 2, 2026.

[3] Zoho Corporation, Creating and Managing Form Workflows – Zoho Creator Help. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-form-workflows/articles/create-new-form-workflow. Accessed Sep. 2, 2026.

[4] Zoho Corporation, Zoho Creator: Best Practices to Build Better Apps. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/getting-started/articles/zoho-creator-best-practices-to-build-better-apps. Accessed Sep. 2, 2026.

[5] J. Dean and S. Ghemawat, “MapReduce: Simplified Data Processing on Large Clusters,” Proceedings of OSDI'04, pp. 137–150, 2004. Available: https://research.google/pubs/mapreduce-simplified-data-processing-on-large-clusters/. Accessed Sep. 2, 2026.

[6] Apache Software Foundation, Apache Hadoop MapReduce Tutorial. Available: https://hadoop.apache.org/docs/current1/mapred_tutorial.html. Accessed Sep. 2, 2026.

[7] Apache Software Foundation, Apache Hadoop Mapper API Documentation. Available: https://hadoop.apache.org/docs/current/api/org/apache/hadoop/mapreduce/Mapper.html. Accessed Sep. 2, 2026.

[8] United Nations Department of Economic and Social Affairs, Sustainable Development Goal 4: Quality Education. Available: https://sdgs.un.org/goals/goal4. Accessed Sep. 2, 2026.

[9] United Nations Department of Economic and Social Affairs, Sustainable Development Goal 9: Industry, Innovation and Infrastructure. Available: https://sdgs.un.org/goals/goal9. Accessed Sep. 2, 2026.

[10] United Nations Department of Economic and Social Affairs, Sustainable Development Goal 12: Responsible Consumption and Production. Available: https://sdgs.un.org/goals/goal12. Accessed Sep. 2, 2026.

[11] Department of Computer Science and Engineering, SIMATS Engineering, CSA15 – Cloud Computing and Big Data Analytics: Library Book Reservation System using Zoho Creator (SaaS) with Big Data Analytics, course assignment document.

[12] Department of Computer Science and Engineering, SIMATS Engineering, Instructions for Common Assignment, Evaluation and Submission, course instruction document.

Submission package: 30-page report PDF, screenshot evidence PDF, sample transaction CSV, mapper.py, reducer.py and MapReduce result CSV.